

Retail Customer Segmentation Analysis

Customer Intelligence Through Unsupervised Machine Learning

Final Analytical Report

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Executive Summary

This project aimed to identify meaningful customer segments within a large-scale retail transaction dataset using unsupervised machine learning techniques. The analysis was conducted on more than one million retail transactions collected between 2009 and 2011 from a UK-based online retailer.

After extensive data preparation and feature engineering, a customer-level analytical dataset was developed using Recency, Frequency, Monetary Value (RFM), Average Order Value, Basket Size, and Customer Lifetime metrics. Multiple clustering algorithms were evaluated, including K-Means, Agglomerative Clustering, and Gaussian Mixture Models.

Based on a comparative evaluation using Silhouette Score, Calinski-Harabasz Index, Davies-Bouldin Index, and business interpretability criteria, K-Means was selected as the final model. The model successfully identified four distinct customer segments representing different purchasing behaviors and business value profiles.

The results revealed a small group of highly valuable loyal customers responsible for substantial revenue generation, a large inactive segment with high churn risk, a specialized group of high-volume buyers, and a core segment of regular customers. These findings provide actionable insights that can support targeted marketing campaigns, customer retention initiatives, and personalized engagement strategies.

The project demonstrates how unsupervised learning techniques can transform transactional data into strategic business intelligence, enabling data-driven decision-making and more effective customer relationship management.

1. Business Objective

The primary objective of this project was to identify meaningful customer segments within a large retail transaction dataset using unsupervised machine learning techniques.

Understanding customer behavior is a critical component of modern business strategy. Organizations that can accurately segment their customers are better positioned to personalize marketing efforts, improve customer retention, optimize resource allocation, and maximize customer lifetime value.

This analysis focuses on customer clustering, aiming to group customers with similar purchasing behaviors based on transactional patterns rather than predefined labels. Since no target variable exists, unsupervised learning provides an effective framework for discovering hidden structures within the data.

The specific business goals of the project were:

- Identify distinct customer groups based on purchasing behavior.
- Compare multiple clustering algorithms and evaluate their effectiveness.
- Select the most appropriate model based on quantitative metrics and business interpretability.
- Generate actionable business insights that support strategic decision-making.
- Provide recommendations for customer engagement, retention, and revenue growth initiatives.

The final outcome is a customer segmentation framework that can help business stakeholders better understand their customer base and develop targeted strategies for each segment.

2. Dataset Description

The analysis was conducted using the Online Retail II dataset, a publicly available transactional dataset frequently used in customer analytics and machine learning applications.

The dataset contains historical online retail transactions from a UK-based non-store retailer between December 2009 and December 2011. The company primarily sells unique gift items and serves both individual and business customers.

After combining both yearly worksheets, the dataset contained approximately 1.07 million transaction records and eight primary variables.

The main attributes included:

Variable	Description
Invoice	Unique transaction identifier
StockCode	Product identifier
Description	Product description

Quantity	Number of items purchased
InvoiceDate	Date and time of transaction
Price	Unit price of the product
Customer ID	Unique customer identifier
Country	Customer location

The dataset provides a rich source of behavioral information that enables customer-level analysis through transaction aggregation. Because the objective of the project was customer segmentation, the transactional data was transformed into customer-centric features representing purchasing behavior over time.

One notable challenge involved missing Customer ID values, which represented approximately 22.8% of the dataset. Since customer identification is essential for segmentation analysis, records without valid customer identifiers were excluded during the feature engineering process.

The size, diversity, and transactional nature of the dataset make it particularly suitable for unsupervised learning applications focused on customer intelligence and behavioral segmentation.

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 07:45:00	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 07:45:00	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 07:45:00	1.25	13085.0	United Kingdom

3. Data Exploration and Preparation

Prior to model development, an exploratory analysis was conducted to understand the structure, quality, and characteristics of the dataset.

After combining both yearly worksheets, the dataset contained 1,067,371 transaction records and eight variables describing customer purchases, products, pricing, and transaction timing.

An initial data quality assessment revealed missing values primarily within the Customer ID field, representing approximately 22.8% of all records. Because customer identification is required to construct customer-level behavioral features, transactions without valid customer identifiers were excluded from the segmentation analysis.

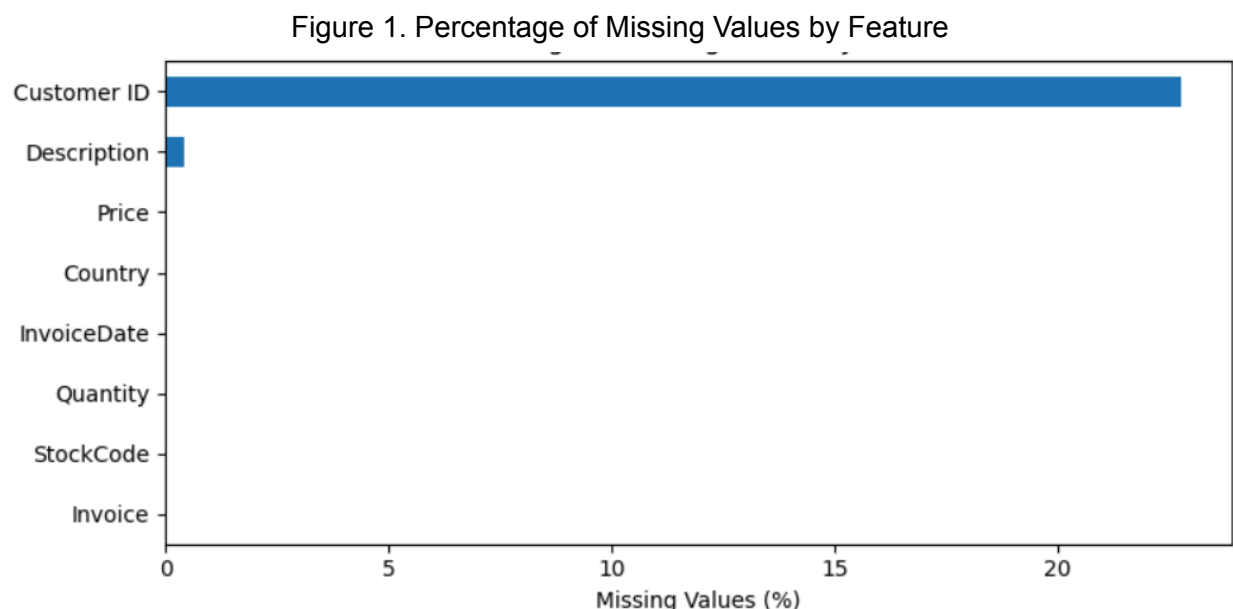
Additional data quality issues were identified during exploration. The dataset contained cancelled transactions, which were indicated by invoice numbers beginning with the letter "C". Since these records represent returns rather than purchases, they were removed from the analytical dataset.

Furthermore, transactions with non-positive quantities or prices were excluded to ensure that customer purchasing behavior was represented accurately.

The data preparation process therefore included:

- Removal of records with missing Customer IDs.
- Removal of cancelled transactions.
- Removal of transactions with non-positive quantities.
- Removal of transactions with non-positive prices.
- Verification of data types and date formats.
- Creation of customer-level aggregated metrics.

These preprocessing steps improved data consistency and ensured that subsequent modeling was based on valid customer purchasing activity.



4. Feature Engineering

The transactional dataset was originally structured at the invoice-item level, where each row represented a product purchased within a transaction. Since customer segmentation requires customer-level analysis, the data was transformed into a customer-centric dataset through feature engineering.

The foundation of this transformation was the RFM framework, a widely used methodology for evaluating customer purchasing behavior.

The following features were generated:

Feature	Description
Recency	Number of days since the customer's most recent purchase
Frequency	Number of completed transactions made by the customer
Monetary	Total revenue generated by the customer
Average Order Value	Average monetary value per transaction
Basket Size	Average quantity of products purchased per transaction
Customer Lifetime	Number of days between the customer's first and last purchase

These variables were designed to capture different dimensions of customer behavior, including purchasing frequency, spending capacity, transaction size, and long-term engagement.

Initial exploratory analysis revealed substantial positive skewness in several variables, particularly Frequency and Monetary Value. Such distributions can negatively affect distance-based clustering algorithms by allowing extreme observations to dominate the clustering process.

To mitigate this issue, logarithmic transformations were applied to the behavioral variables. The transformation reduced skewness, improved distributional balance, and enhanced cluster separability.

Finally, all variables were standardized using StandardScaler to ensure that each feature contributed equally during model training.

The resulting analytical dataset provided a robust representation of customer purchasing behavior suitable for unsupervised learning applications

Figure 2. Distribution of Monetary Before Log Transformation

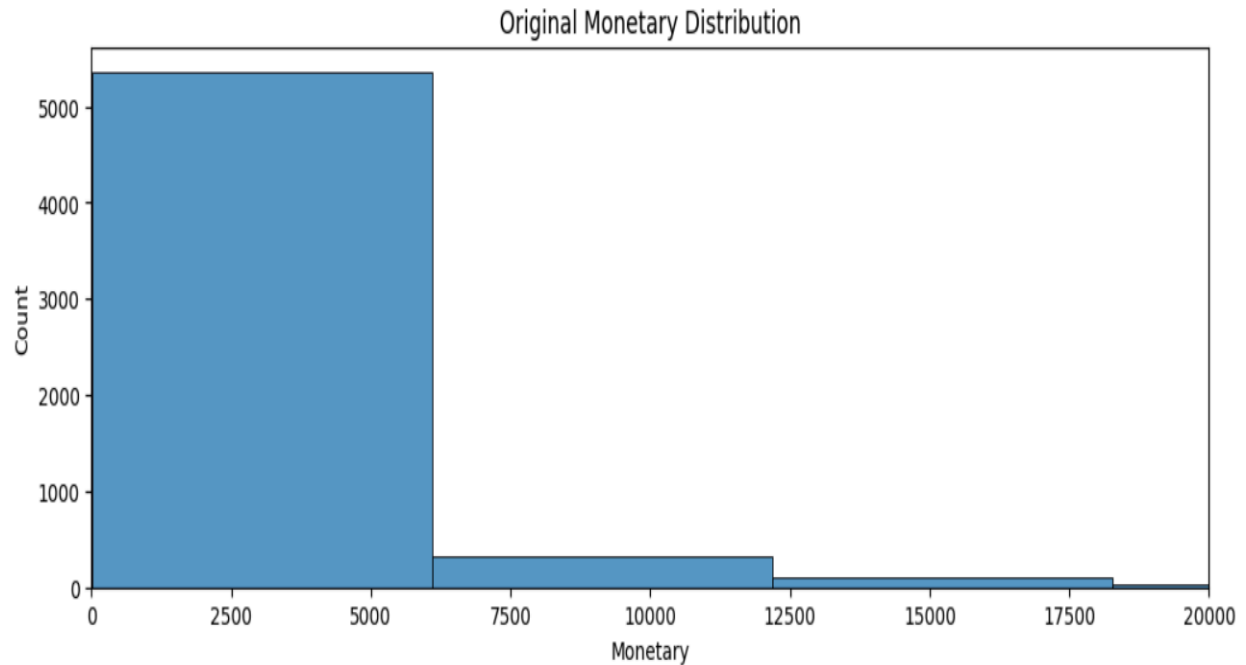
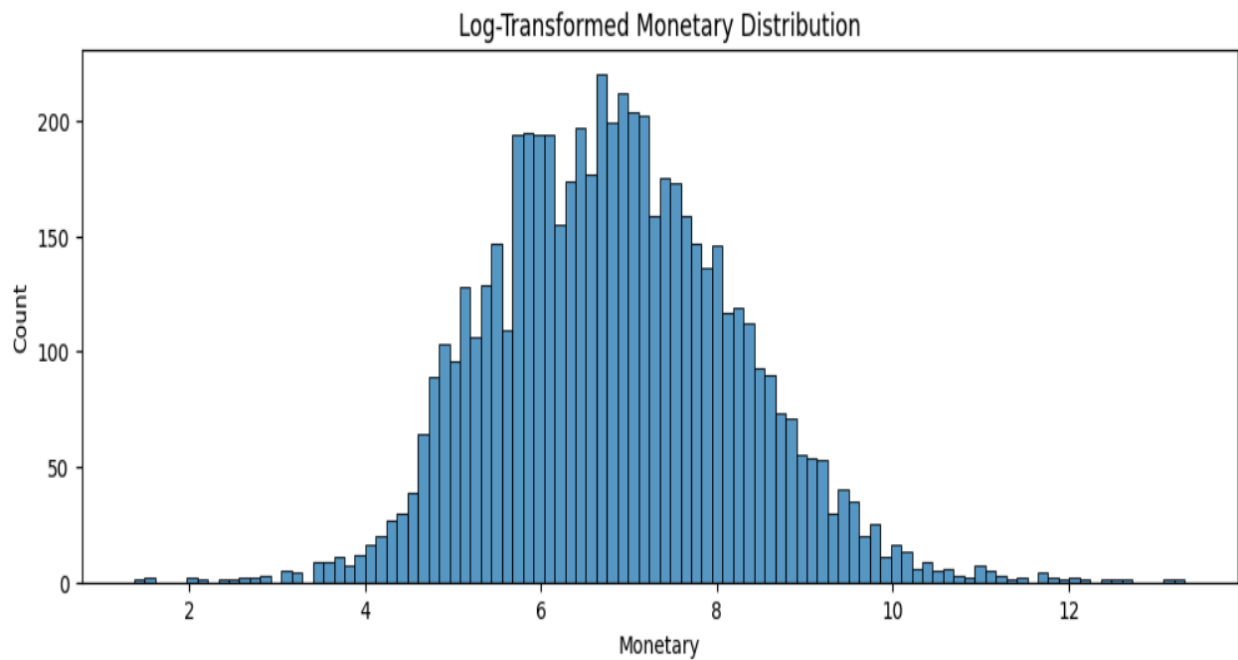


Figure 3. Distribution of Monetary After Log Transformation



5. Model Development and Comparison

To identify meaningful customer segments, multiple unsupervised learning algorithms were evaluated and compared. The objective was not only to maximize clustering performance

metrics but also to ensure that the resulting segments were interpretable and actionable from a business perspective.

Prior to model training, the engineered customer-level features were standardized using StandardScaler. Principal Component Analysis (PCA) was then applied as an exploratory visualization technique to assess the separability of potential customer groups in a lower-dimensional space.

Three clustering approaches were evaluated:

K-Means Clustering

K-Means partitions observations into a predefined number of clusters by minimizing the within-cluster sum of squares. Due to its simplicity, scalability, and interpretability, K-Means is one of the most widely used customer segmentation techniques.

Agglomerative Clustering

Agglomerative Clustering is a hierarchical clustering method that progressively merges observations based on similarity. Unlike K-Means, it does not assume spherical cluster structures and can capture more complex relationships between observations.

Gaussian Mixture Models (GMM)

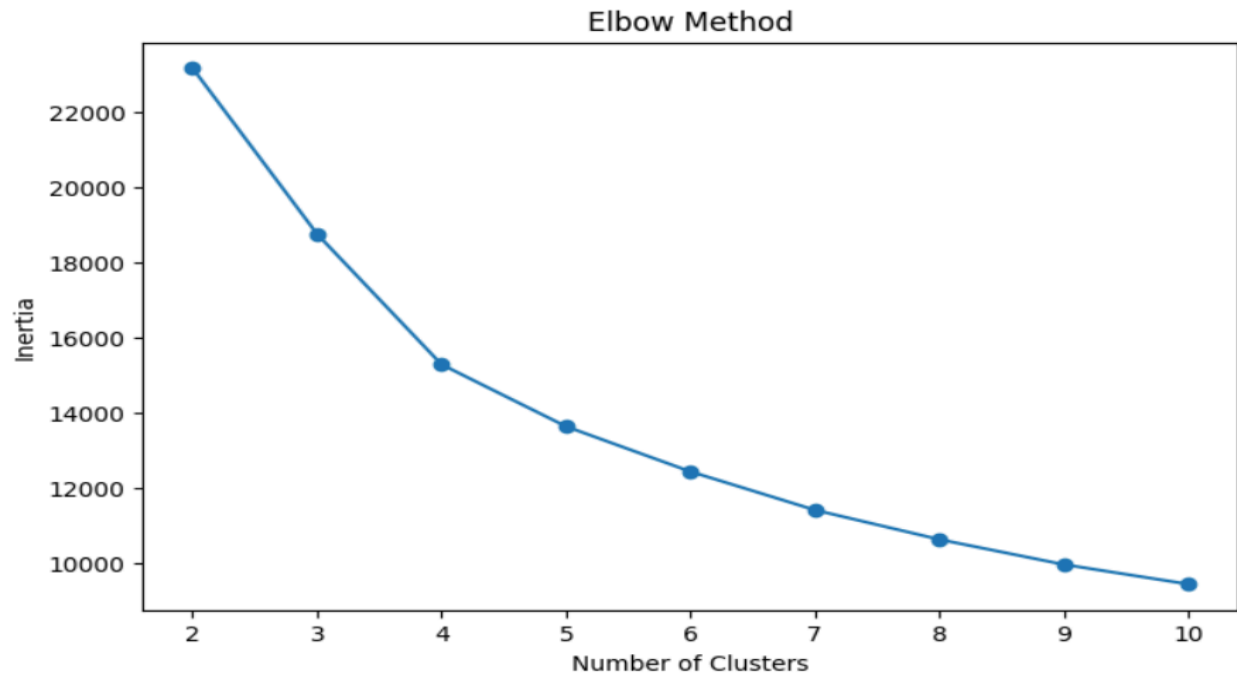
Gaussian Mixture Models represent the data as a mixture of probability distributions. This probabilistic approach allows observations to belong partially to multiple clusters and can model more flexible cluster shapes than K-Means.

To evaluate model performance, three complementary clustering metrics were used:

- Silhouette Score (higher is better)
- Calinski-Harabasz Index (higher is better)
- Davies-Bouldin Index (lower is better)

These metrics provide different perspectives on cluster quality, including cohesion, separation, and overall cluster structure.

Figure 4. Elbow Method for Determining the Optimal Number of Clusters



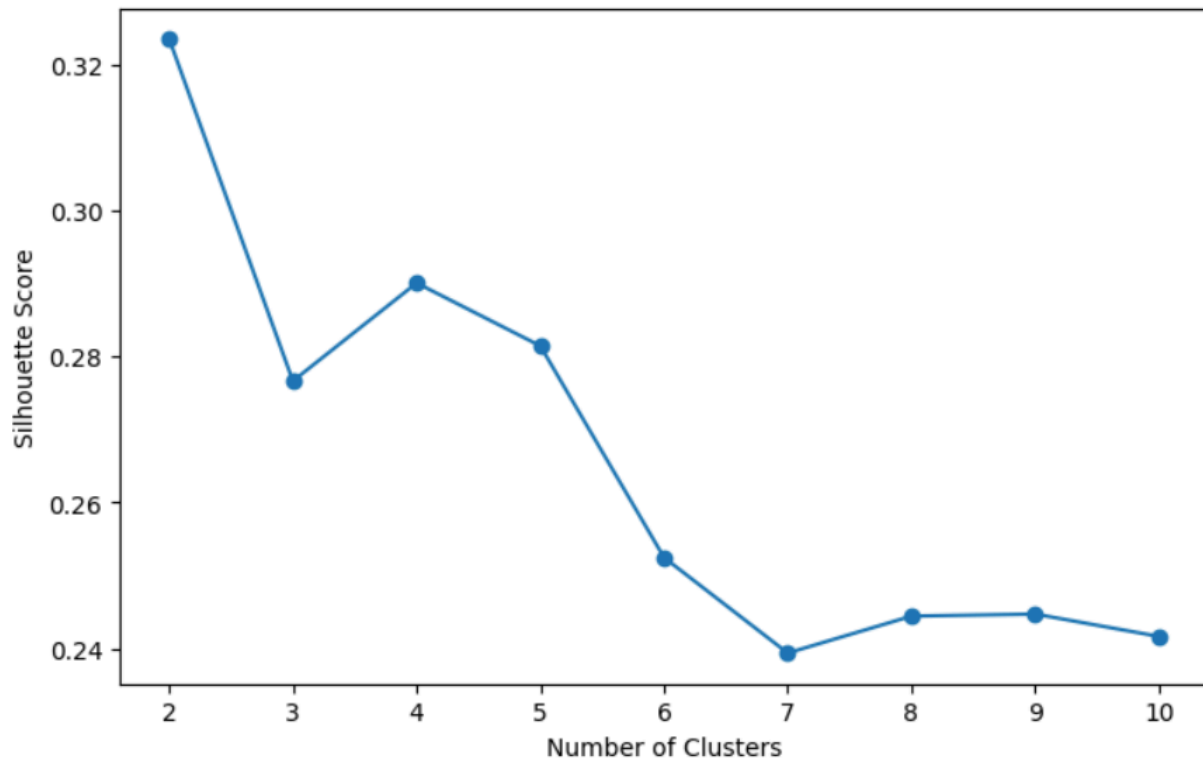
The Elbow Method suggested that cluster quality improvements began to diminish after four clusters, providing an initial indication that a four-cluster solution could balance model complexity and interpretability.

Table 2. Clustering Model Performance Comparison

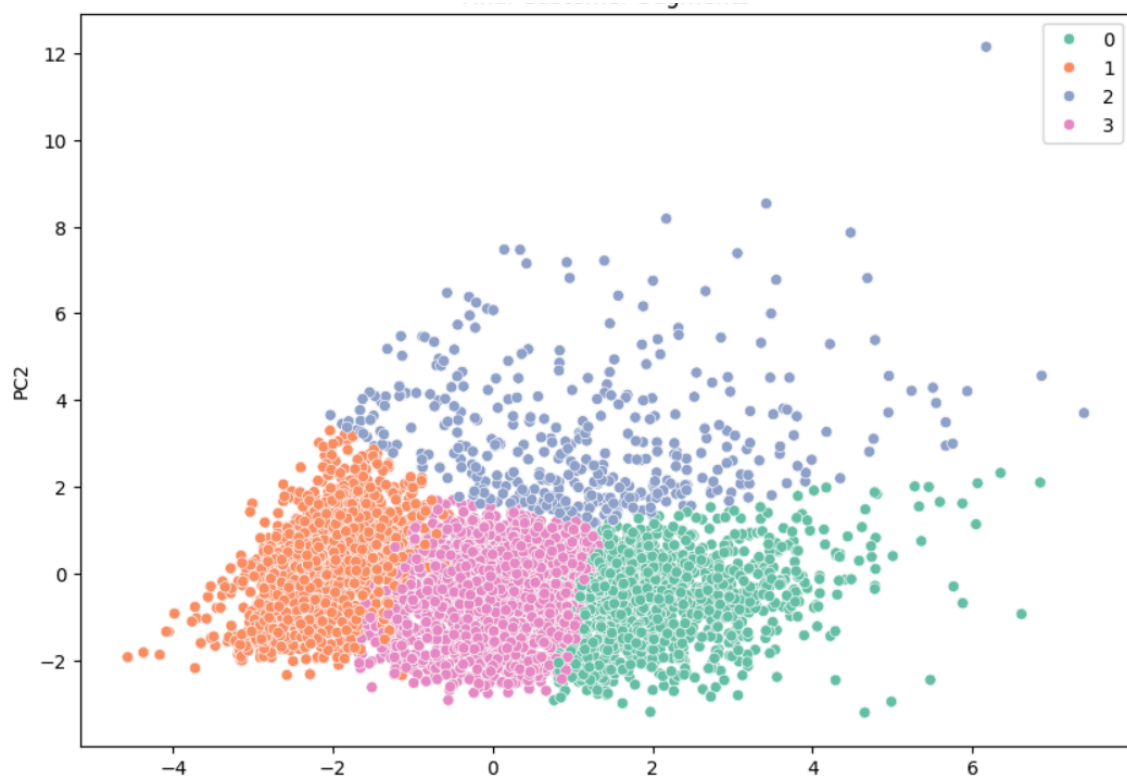
	Model	Silhouette Score	Calinski-Harabasz	Davies-Bouldin
0	KMeans	0.290116	2560.133805	1.163931
1	Agglomerative	0.260756	1996.736476	1.130575
2	Gaussian Mixture	0.162694	1370.285207	1.955225

As shown in Table 2, K-Means achieved the highest Silhouette Score and the highest Calinski-Harabasz Index, indicating stronger cluster separation and internal cohesion compared to the alternative approaches. Although Agglomerative Clustering achieved a slightly lower Davies-Bouldin score, the overall performance and interpretability of K-Means made it the most suitable candidate for customer segmentation.

Figure 5. Silhouette Scores Across Different Numbers of Clusters



The Silhouette Analysis provided additional evidence for selecting four clusters. While the Elbow Method suggested a potential solution between three and four clusters, the Silhouette Score reached its highest value at $k=4$...



PCA visualization provides a two-dimensional representation of customer behavior patterns and reveals the existence of distinct customer groups within the dataset. While some overlap remains expected due to the complexity of real-world purchasing behavior, the segmentation structure supports the validity of the clustering approach.

6. Model Selection

Following the comparative evaluation of the clustering algorithms, K-Means was selected as the final customer segmentation model.

The selection process considered both quantitative performance metrics and business interpretability. While all three models successfully identified customer groupings, K-Means consistently demonstrated superior overall performance.

Table 2 showed that K-Means achieved the highest Silhouette Score (0.290) and the highest Calinski-Harabasz Index (2560.13), indicating stronger cluster cohesion and better separation between groups. Although Agglomerative Clustering produced a slightly lower Davies-Bouldin Index, its overall cluster structure was less distinct and less interpretable from a business perspective.

Gaussian Mixture Models produced the weakest performance across the evaluation metrics, suggesting that the probabilistic approach did not adequately capture the underlying structure of the customer behavior data.

Beyond statistical performance, K-Means generated customer segments that were highly interpretable and aligned with recognizable business profiles. The resulting clusters revealed distinct behavioral patterns that were interpreted as high-value customers, inactive customers, bulk buyers, and regular customers based on their purchasing characteristics, enabling practical business applications and strategic decision-making.

For these reasons, K-Means with four clusters was selected as the final model for customer segmentation.

7. Customer Segment Analysis

Customer segmentation was performed using the selected K-Means model with four clusters. The resulting groups exhibit distinct purchasing behaviors and provide valuable insights into customer engagement, spending patterns, and long-term value.

To facilitate interpretation, the average behavioral characteristics of each cluster were analyzed and compared.

Figure 7. Customer Segment Heatmap

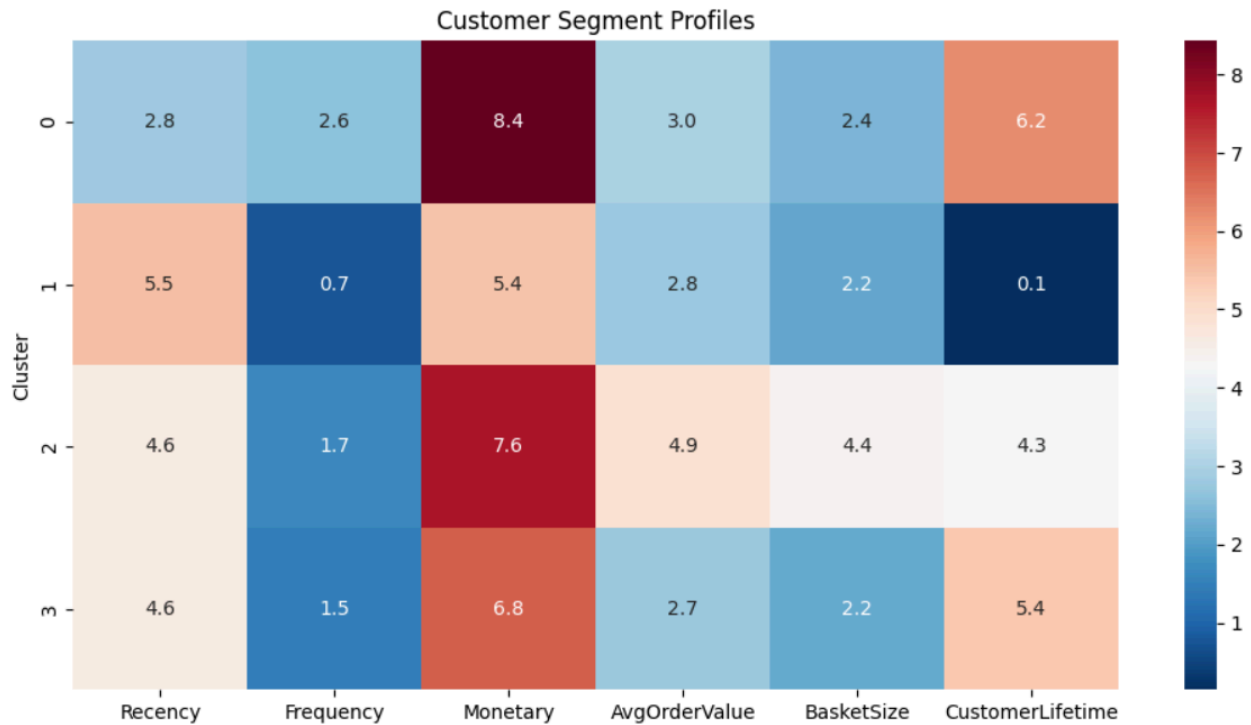


Figure 7 illustrates the behavioral differences across customer segments. Significant variation can be observed in recency, purchasing frequency, monetary contribution, transaction size, and customer lifetime, confirming the effectiveness of the segmentation process.

Table 3. Average Behavioral Characteristics by Customer Segment

	Recency	Frequency	Monetary	AvgOrderValue	BasketSize	CustomerLifetime
Cluster						
0	15.53	12.68	4579.84	18.40	10.20	504.69
1	240.54	1.08	229.78	15.36	7.76	0.15
2	101.97	4.26	2075.60	134.95	76.84	69.54
3	102.32	3.47	873.89	14.47	7.97	220.74

Segment Interpretation

Cluster 0 — VIP Loyal Customers

This segment contains highly valuable and loyal customers characterized by frequent purchases, recent activity, high spending levels, and long customer lifetimes. These customers represent the most important revenue-generating segment and should be prioritized for retention and loyalty initiatives.

Cluster 1 — One-Time / Churned Customers

Customers in this segment exhibit very low purchasing activity and long periods of inactivity. Their low monetary contribution and limited engagement indicate a high probability of churn. Re-engagement campaigns may be required to recover value from this group.

Cluster 2 — High-Value Bulk Buyers

This segment generates substantial revenue through large individual purchases and exceptionally large basket sizes. Their behavior suggests wholesale purchasing patterns or infrequent but high-value transactions. Personalized account management strategies may maximize value from this segment.

Cluster 3 — Regular Customers

These customers demonstrate moderate spending, recurring purchasing behavior, and longer-term engagement. Although they do not generate the same value as VIP customers, they represent a stable customer base that contributes consistently to business performance.

8. Key Business Insights

The customer segmentation analysis revealed four distinct customer groups with significantly different purchasing behaviors and business value.

Insight 1: A Small Segment Generates Disproportionate Value

Cluster 0 (VIP Loyal Customers) represents the most valuable customer group. These customers exhibit the highest purchase frequency, strongest monetary contribution, and longest customer lifetime. Although they represent only a portion of the customer base, they generate a disproportionate share of overall revenue and should be prioritized for retention programs and personalized marketing initiatives.

Insight 2: Customer Churn Represents a Significant Opportunity

Cluster 1 consists of customers with very low engagement and extended inactivity periods. Their limited purchase history suggests a high probability of churn. Targeted reactivation campaigns

could help recover revenue from this segment at a relatively low acquisition cost compared to attracting entirely new customers.

Insight 3: Bulk Buyers Require Specialized Strategies

Cluster 2 demonstrates exceptionally high average order values and basket sizes, indicating purchasing patterns that differ substantially from the rest of the customer base. These customers may benefit from dedicated account management, volume discounts, or tailored product recommendations designed for large transactions.

Insight 4: Regular Customers Represent Growth Potential

Cluster 3 contains customers with moderate spending levels and recurring purchasing activity. This segment provides stable revenue and presents opportunities for upselling and cross-selling initiatives that could increase their long-term value.

Business Recommendations

Based on the segmentation results, the following actions are recommended:

- Develop loyalty and retention programs for VIP customers.
- Implement re-engagement campaigns targeting inactive customers.
- Create personalized offerings for bulk buyers.
- Design upselling strategies for regular customers.
- Incorporate customer segmentation into future marketing and customer relationship management initiatives.

9. Limitations and Future Work

While the project successfully identified meaningful customer segments, several limitations should be acknowledged.

First, the analysis relied exclusively on transactional purchase data. Additional customer information such as demographics, acquisition channels, geographic characteristics, or marketing interactions could provide a more comprehensive understanding of customer behavior.

Second, the segmentation was based on historical purchasing patterns and assumes that future customer behavior will remain relatively stable. Changes in customer preferences or market conditions may alter segment characteristics over time.

Third, K-Means clustering assumes relatively spherical cluster structures and may not fully capture more complex relationships within the data. Alternative clustering approaches could reveal additional behavioral patterns.

Future work could include:

- Incorporating demographic and marketing data into the segmentation process.
- Testing advanced clustering techniques such as DBSCAN, HDBSCAN, or Spectral Clustering.
- Developing predictive models to estimate customer lifetime value and churn probability.
- Implementing automated customer segment monitoring to detect behavioral changes over time.
- Evaluating segmentation stability across different time periods and business cycles.

These improvements could further enhance the analytical value and business applicability of the segmentation framework.

10. Conclusion

This project demonstrated the application of unsupervised machine learning techniques to customer segmentation using transactional retail data.

After extensive data preparation, feature engineering, dimensionality reduction, and model evaluation, K-Means clustering with four customer segments was selected as the most effective solution. The resulting segmentation successfully identified meaningful differences in customer purchasing behavior, spending patterns, engagement levels, and long-term value.

The analysis revealed the presence of highly valuable loyal customers, inactive customers with elevated churn risk, high-value bulk buyers, and a stable segment of regular customers. These insights provide actionable information that can support customer retention strategies, personalized marketing initiatives, and more efficient resource allocation.

Beyond its analytical findings, the project demonstrates a complete end-to-end data science workflow, including data preparation, feature engineering, model comparison, model selection, business interpretation, and strategic recommendation development.

The proposed segmentation framework provides a practical foundation for data-driven customer relationship management and future analytical initiatives.

